$\Rightarrow ((X^{-1}-I)B)B^{-1} = A^{-1}B^{-1}$

$\Rightarrow (X^{-1}-I)(BB^{-1}) = A^{-1}B^{-1}$ (Theorem 2(a))

$\Rightarrow (X^{-1}-I)I = A^{-1}B^{-1}$ (By the defn. of inverse and part (a))

$\Rightarrow (X^{-1}-I) = A^{-1}B^{-1}$ (Theorem 2(e))

$\Rightarrow (X^{-1}-I)+I = A^{-1}B^{-1}+I$

$\Rightarrow X^{-1}+(-I+I) = A^{-1}B^{-1}+I$ (Theorem 1(b))

$-I+I = [-i_1\ -i_2 \ldots\ -i_n] + [i_1\ i_2 \ldots\ i_n]$

$= [-i_1+i_1\ -i_2+i_2 \ldots\ -i_n+i_n]$ (Vector addition)

$= [0\ 0 \ldots\ 0]$ (Algebraic property iv)

$= 0$

$\Rightarrow X^{-1}+0 = A^{-1}B^{-1}+I \Rightarrow X^{-1} = A^{-1}B^{-1}+I$ (Theorem 1(c))

$(X^{-1})^{-1} = X = (A^{-1}B^{-1}+I)^{-1}$ (Theorem 6(a))

$\boxed{\therefore X = (A^{-1}B^{-1}+I)^{-1}}$ □

21) Explain why the columns of an $n\times n$ matrix A are linearly independent when A is invertible.

Solution: By Theorem 5, for any arbitrary b in $\mathbb{R}^n$, $Ax=b$ has a unique solution. $\therefore$ The matrix equation $Ax=0$ has a unique solution.

By Defn. of Ax, $x_1a_1+x_2a_2+\ldots\ldots+x_na_n=0$, where $A=[a_1\ a_2 \ldots\ldots a_n]$.

The trivial solution $x_1=x_2=\ldots\ldots=x_n=0$ is the only soln. of this equation. $\therefore$ By Defn, $a_1, a_2, \ldots, a_n$ are linearly independent □

22) Explain why the columns of an $n\times n$ matrix A span $\mathbb{R}^n$ when A is invertible.

Solution: By Theorem 5, the matrix equation $Ax=b$ has a unique solution for any $b\in\mathbb{R}^n$, if A is invertible. By Theorem 4 in Chapter 1, the columns of A span $\mathbb{R}^n$. □

23) Suppose A is $n\times n$ and the equation $Ax=0$ has only the trivial solution. Explain why A has n pivot columns and A is row equivalent to I_n. By Theorem 7, this shows that A must be invertible. (This exercise and Exercise 24 will be cited in Section 2.3)